α-$BaCu_2Se_2$ Thin-Film XRD Analysis and Bayesian Optimization

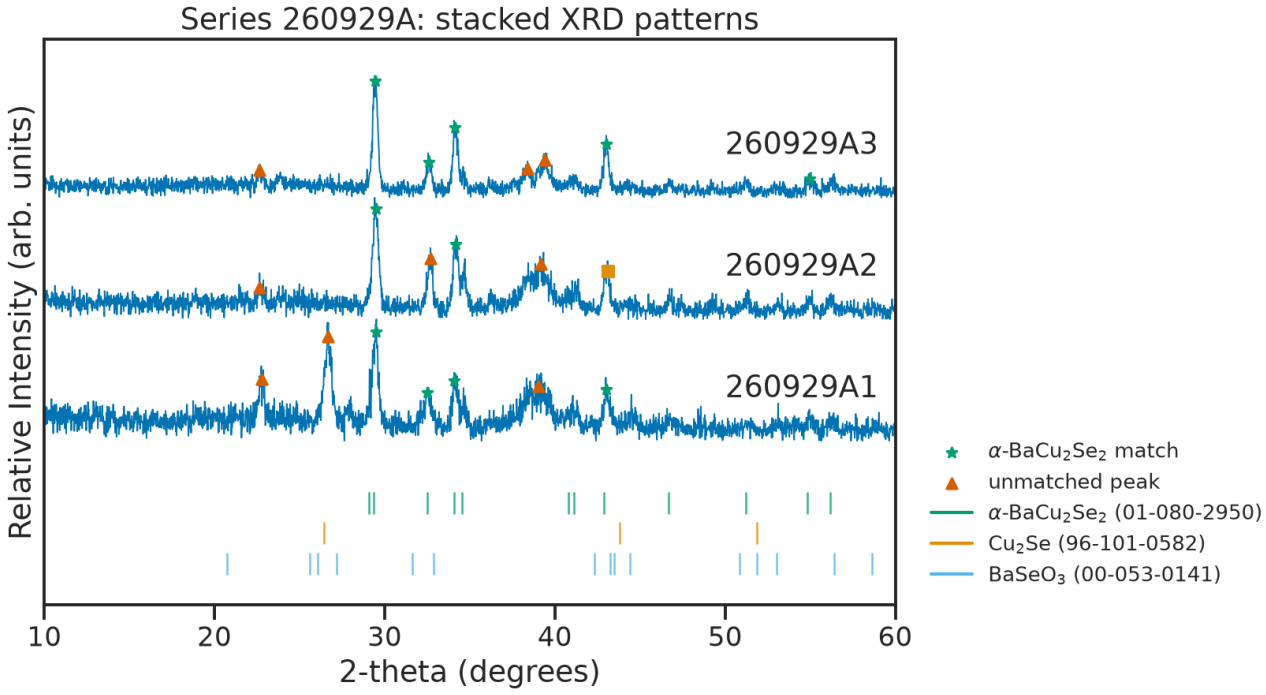
NOMAD-driven data extraction, phase matching, and GP-based synthesis planning

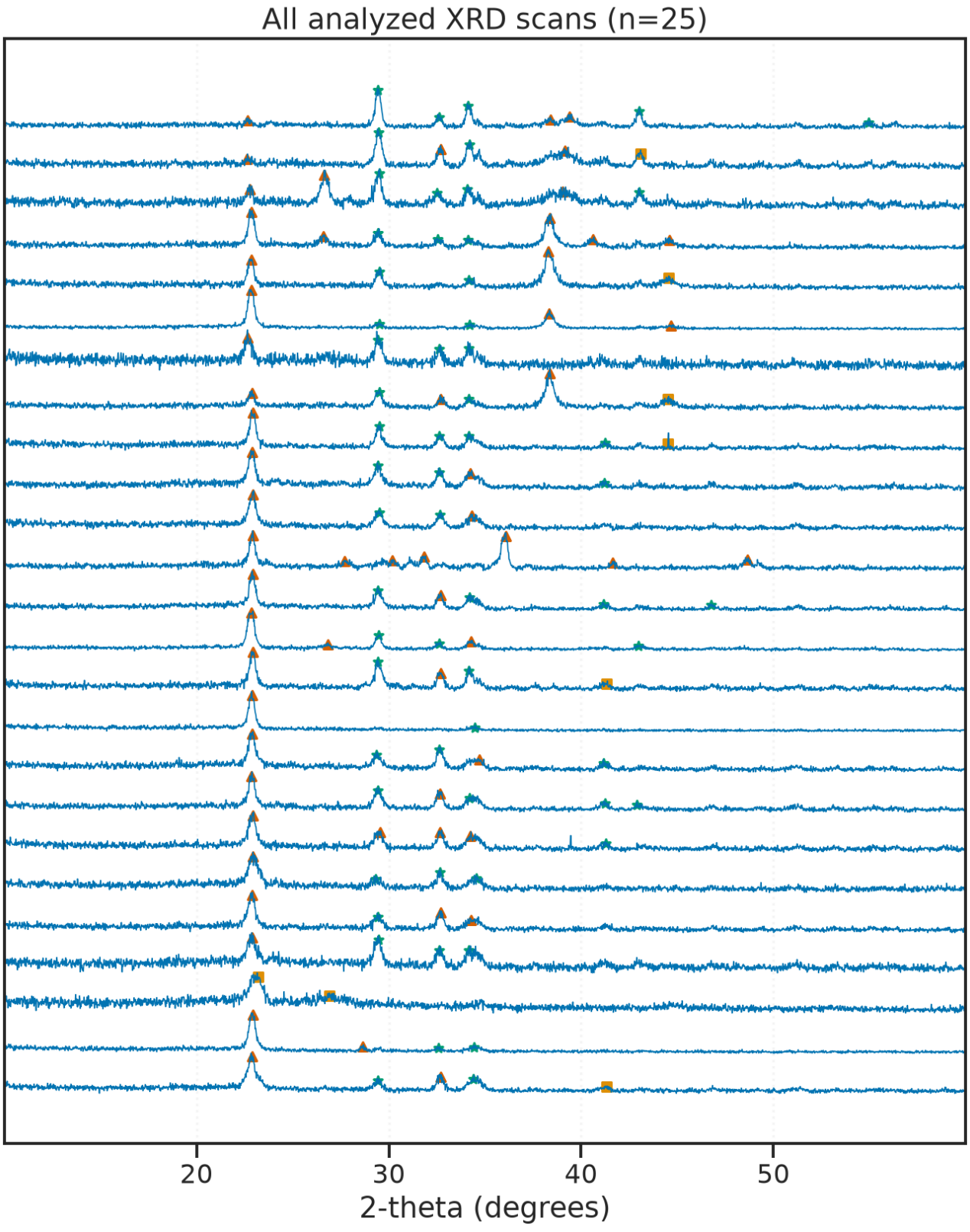
1. What was done

Synthesis records and XRD scans were pulled directly from the NOMAD Oasis REST API for the 42 α-$BaCu_2Se_2$ thin-film samples (INLThinFilmStack entries). Synthesis parameters (spin coating, soft/hard bake, selenization) and selenium-pulse settings were extracted for all samples; 35 include pulse on/off data. Each record also carries the precursor Cu/(Ba+Cu) ratio, the number of spin-coating cycles (deposited layers, excluding dynamic-dispense/seed-layer preparation steps), the Mo-coating flag, and the spin-to-selenization day gap. For the 25 films with XRD scans, diffraction patterns were matched against reference cards for the α-$BaCu_2Se_2$ target (01-080-2950), Cu_2Se (96-101-0582; α-Cu_2Se), and $BaSeO_3$ (00-053-0141). Phase purity and crystallinity (1/FWHM of the strongest target match) were computed per film. A Gaussian-process Bayesian optimizer was trained on the 21 complete films to recommend the next synthesis conditions, optimizing objective = (0.7 × phase purity + 0.3 × normalized crystallinity) × (1 – 0.25 × A) over a five-input space: hard-bake temperature, selenization temperature and time, Cu/(Ba+Cu) ratio, and one categorical Se pulse profile derived from on/off timings. CBC is defined as $n_{Cu}/(n_{Ba}+n_{Cu})$; the candidate grid is 0.63–0.71, so a ratio of 1 (no Ba) is excluded. A is the unmatched/overlap share of detected peak-interval area, a screening proxy—not a quantitative phase fraction or a confirmed assignment of the ~23° feature.

2. XRD phase identification

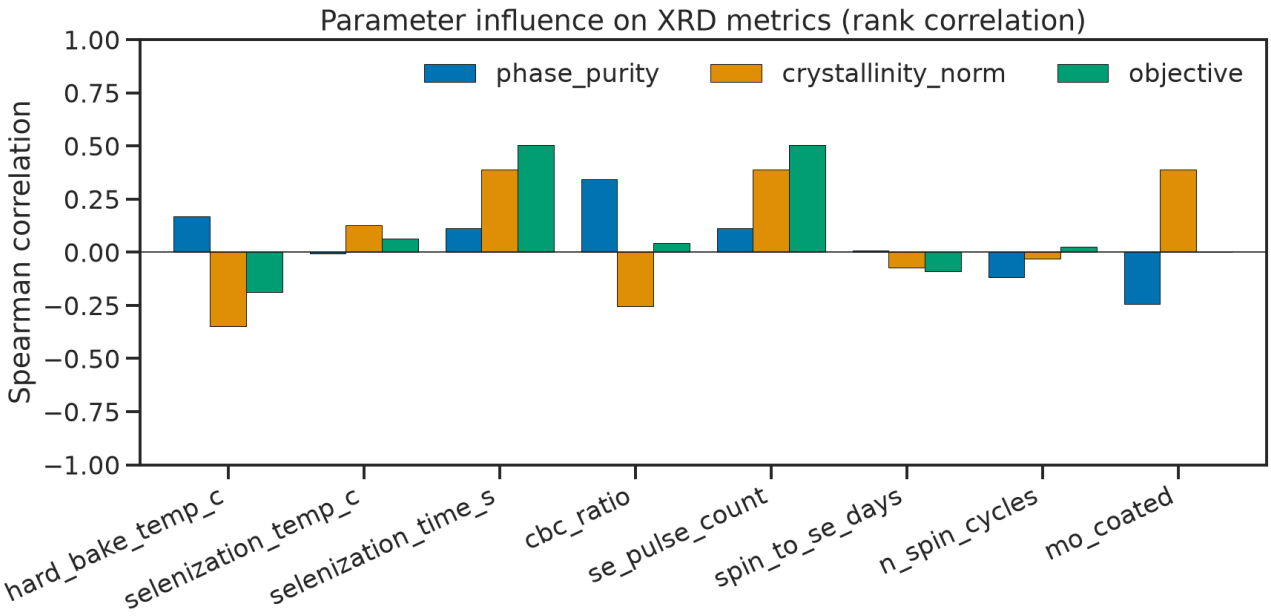
Stacked XRD patterns per series with reference bars for the target and impurity phases. Markers flag target (star), impurity (square), and unmatched (triangle) peaks.

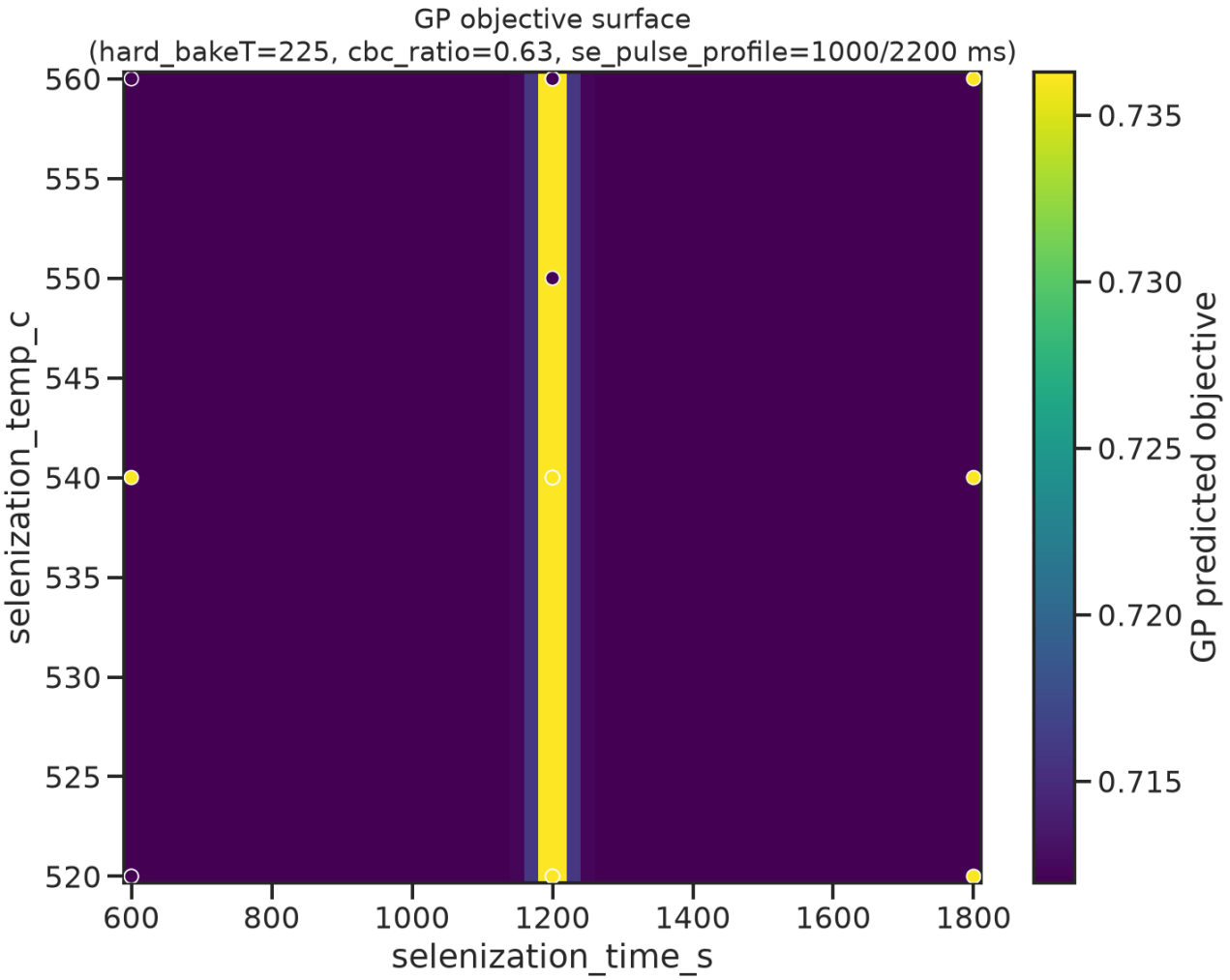




3. Main results

Across the analyzed films the mean phase purity is 0.94 and mean crystallinity is 2.18. The best film under the penalized score is 260929A3 (score 0.91; unpenalized score 0.99; purity 1.00, crystallinity 2.83, A 0.33). The optimizer uses the five selected inputs only. Other process and substrate descriptors remain available for descriptive exploration; the pulse profile is categorical and is not treated as an ordinal correlation.



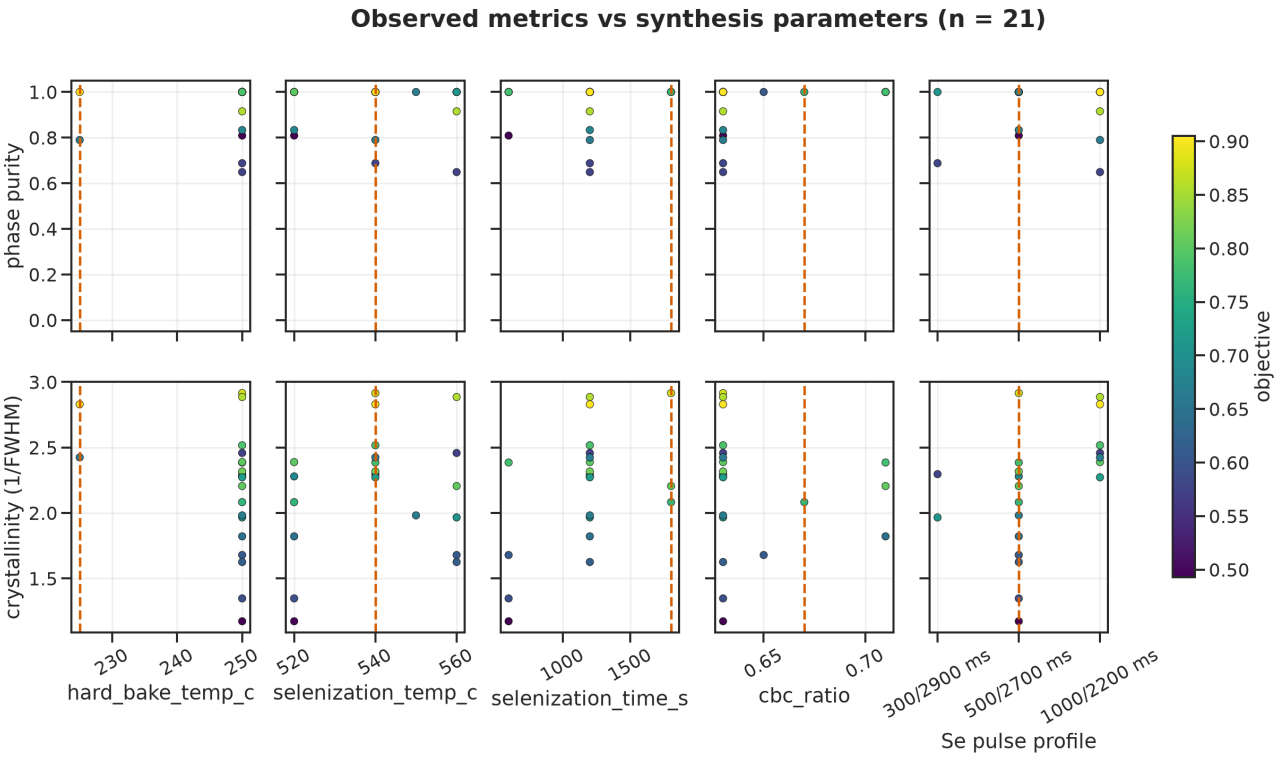


4. Suggested next experiments

Top Gaussian-process suggestions (predicted objective ± std), ranked by expected improvement:

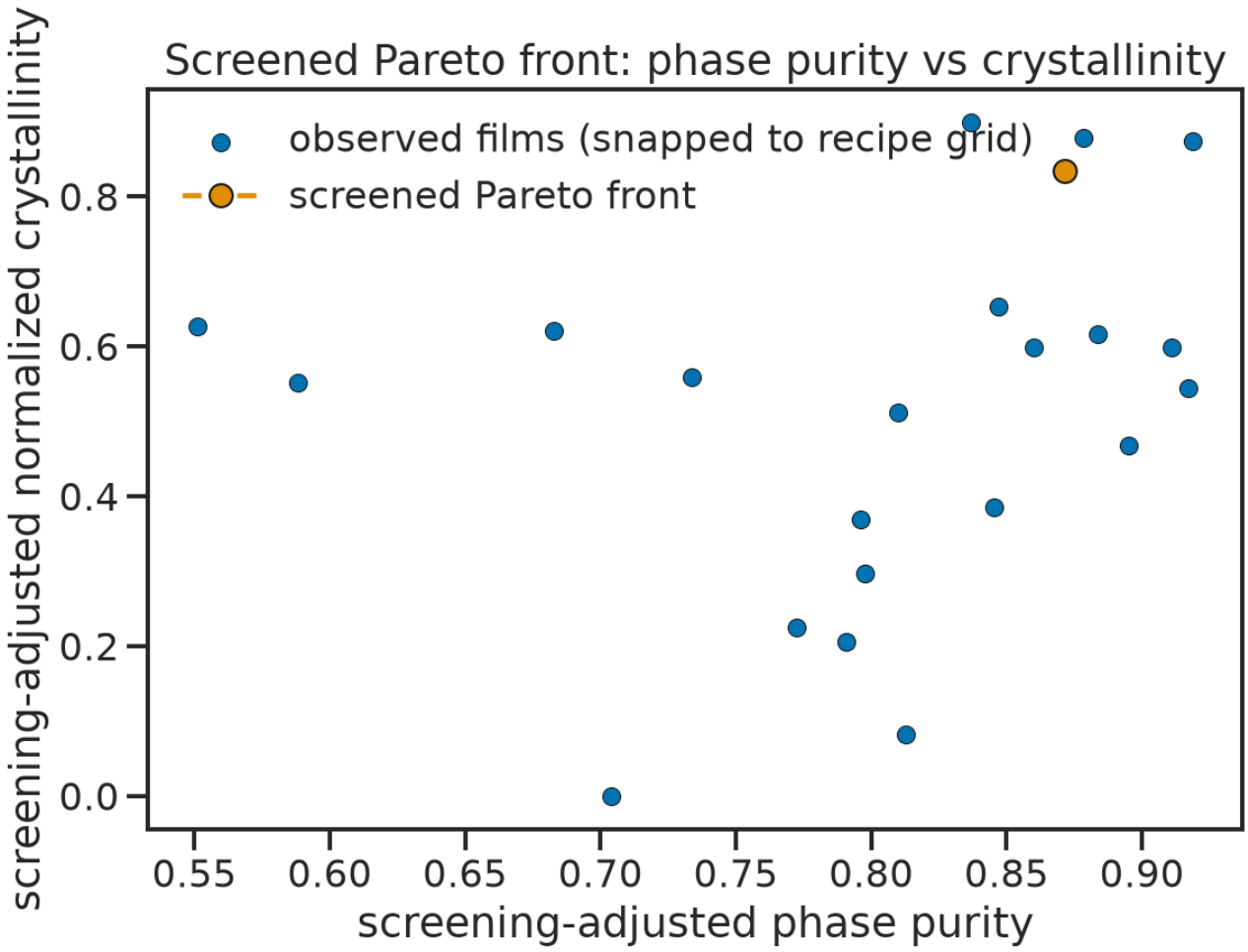
#	Hard bake °C	Selen. temp °C	Selen. time s	CBC ratio	Se pulse profile	Se on ms	Se off ms	Predicted score	GP std
1	225	540	1800	0.670	500/2700 ms	500	2700	0.767	0.051
2	225	540	1800	0.650	500/2700 ms	500	2700	0.767	0.051
3	250	540	1800	0.710	500/2700 ms	500	2700	0.767	0.051
4	225	540	1800	0.670	1000/2200 ms	1000	2200	0.712	0.062
5	250	540	1800	0.710	1000/2200 ms	1000	2200	0.712	0.062

Recommended first experiment: hard bake 225 °C, selenization 540 °C for 1800 s, CBC ratio 0.670, and Se pulse 500/2700 ms profile (500 ms on / 2700 ms off) (predicted objective 0.77).



5. Multi-objective optimization (Ax)

A complementary multi-objective search treats screening-adjusted phase purity and normalized crystallinity as two separate objectives: both are multiplied by the same (1 – 0.25 × A) factor. Ax (BoTorch) fits a Gaussian process per metric and finds the Pareto front via qNEHVI hypervolume maximization. Its five candidate inputs match notebook 3: hard-bake temperature, selenization temperature/time, CBC ratio, and a categorical joint pulse profile. Numeric recipes use the same explicit experiment-scale grids, including 25 °C bake steps, and the profile is reported with exact on/off timings. This remains a Pareto formulation, not the same scalar objective as notebook 3. A is an exploratory detected-peak-area screening proxy, not a phase fraction or a confirmed assignment of the ~23° feature.



#	Hard bake °C	Selen. temp °C	Selen. time s	CBC ratio	Se pulse profile	Se on ms	Se off ms
1	225	540	1800	0.630	500/2700 ms	500	2700
2	250	530	1800	0.630	500/2700 ms	500	2700
3	225	520	1500	0.640	1000/2200 ms	1000	2200
4	250	530	600	0.650	1000/2200 ms	1000	2200
5	250	530	600	0.710	1000/2200 ms	1000	2200